Hamish Huggard

PHYSICS TEACHER & AI ENGINEER

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Physics teacher at Glenfield College, specialising in building **science simulations** and **classroom software**.

M.Sc. in Computer Science (awarded **Best Thesis**) and a conjoint B.Sc. in Physics & Computer Science / B.A. in Philosophy & Mathematics, with **three machine learning publications**. Before teaching I spent five years as a **data scientist and machine learning engineer**, working on clinical AI at Orion Health, research tools at Litmaps, and frontier AI evaluations at METR. In between, I spent a year making educational animations — one hit **3 million views** and led to a **six-figure grant** to make more.

Experience

2026-Present **Glenfield College** — Physics Teacher (Trainee)
Teaching Year 9 and 10 science and Year 12 and 13 physics full-time.
Built my own software suite for teaching, combining slide decks, lesson plans, calendars, timetables, roll call, flashcards, student progress tracking, and interactive simulations.
Coach the robotics club and help run the adventure club.

2024-2025 **Equistamp** — Research Engineer (Contract)
Expert human baseline for benchmarking frontier AI agents on programming and ML tasks, plus QA on model outputs and code for the evaluation suites.

2022-2023 **Global Problems Cheatsheet** — Writer & Animator
Spent a year writing and animating an illustrated primer on the world's most pressing problems — global poverty, pandemics, nuclear war, AI, and climate change.

2020-2021 **Litmaps** — Data Scientist
Built the interactive research-graph visualizations at the core of the product in React and D3.js, and deployed the company's first production AI features.

2018-2020 **Orion Health** — Data Scientist (Graduate)
Developed a novel feature engineering library for biomedical text, improving the F1-score of deep learning models for named entity recognition in clinical notes.

2015-2018 **University of Auckland** — Teaching Assistant
Taught physics and computer science labs alongside my undergraduate study.

Education

2018-2020 *University of Auckland* — **GPA: 4.0 / 4.0**
Master of Science (M.Sc.) — Computer Science
My external examiner described it as **"the best Masters thesis I have seen."**
Best M.Sc. Thesis Award, CS Dept (2020) · Precision Driven Health Scholarship (2019)
Postgraduate Honours Scholarship (2017)

2014-2017 *University of Auckland* — **GPA: 3.81 / 4.0**
Bachelor of Science (B.Sc.) — Physics & Computer Science (double major)

2014-2017 *University of Auckland* — **GPA: 3.81 / 4.0**
Bachelor of Arts (B.A.) — Philosophy & Mathematics (double major)
Senior Scholar Award, Faculty of Arts (2017) · Sir Robert Jones Scholarship in Philosophy (2015)

PUBLICATIONS

2020 *Concept drift detection for medical triage.*
H. Huggard, Y.S. Koh, G. Dobbie, E. Zhang.
ACM SIGIR.

2019 *Feature importance for biomedical named entity recognition.*
H. Huggard, A. Zhang, E. Zhang, Y.S. Koh.
AI: Advances in Artificial Intelligence.

2018 *Predicting air quality from low-cost sensor measurements.*
H. Huggard, Y.S. Koh, P. Riddle, G. Olivares.
Australasian Conference on Data Mining.

TECHNICAL SKILLS

Teaching: physics & mathematics, interactive simulation design, explanatory animation, technical writing.

AI & ML: Python, PyTorch, LLMs, AI Agents, Claude Code, Cursor, NLP, Scikit-Learn, Pandas.

Web & Viz: JavaScript, HTML, D3.js, React, Plotly, CSS.

Engineering: Git, SQL, Postgres, Docker, Linux, Kubernetes.

ADVENTURES

2026 — 64km in 6:54
South Island Ultra-marathon

2023 — 280km in 3 weeks
Camino de Santiago, Portugal

2021 — 3,000km in 6 months
Te Araroa Trail, the length of NZ